

Will My Money Last?

Retirement income Monte Carlo simulator | methodology and results | Leonardo Gama | June 2026

The question

Retirement clients ask one thing before anything else: will my money last? Because the order of yearly returns changes the outcome, this tool runs 10,000 market histories and reports how often the money lasts, with the full range of balances behind it.

Method

Everything is in real (inflation-adjusted) dollars, so spending stays flat in today's money and inflation is not modeled separately. Each simulated year withdraws the planned spending first, then grows what is left by a random real return drawn for that year. Yearly returns are lognormal, so a return is never worse than -100%. A path fails the first year it cannot fund a full withdrawal. Running this 10,000 times, including the unlucky paths where weak markets land early in retirement, is how sequence-of-returns risk shows up in the result.

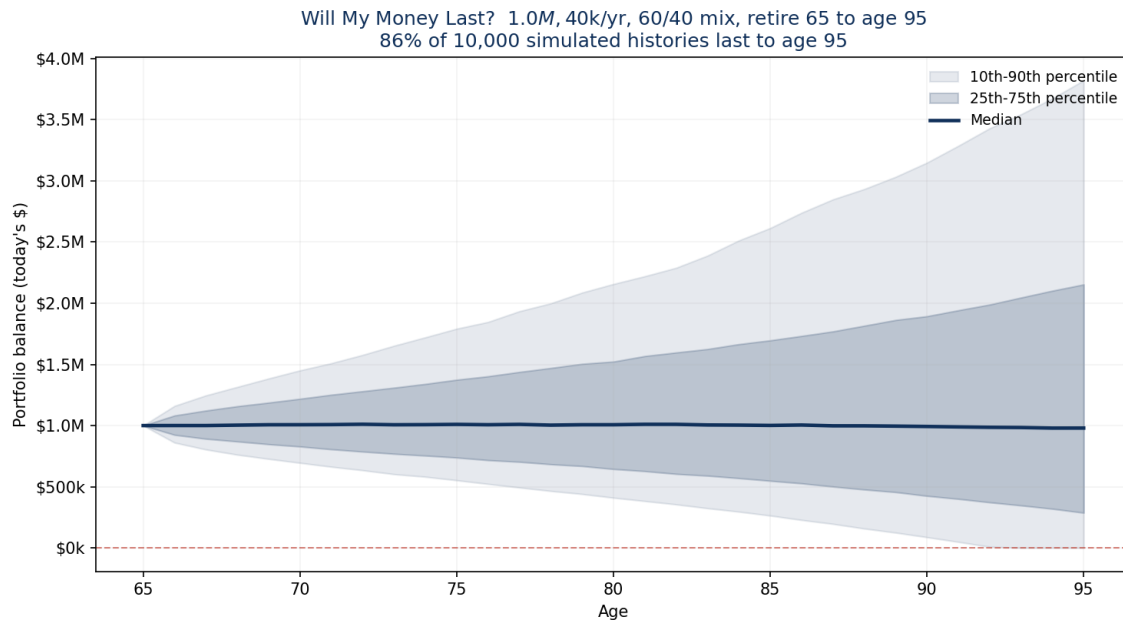
Asset	Real return (annual)	Volatility
Stocks	7%	20%
Bonds	2%	6%

Long-run US history (SBBI/Ibbotson 1926 to 2024, cross-checked against Damodaran). Stocks at 7% is set a touch below the historical arithmetic average on purpose, so the tool does not overstate the odds. Stocks and bonds are assumed uncorrelated.

Results

How much you spend relative to savings is the biggest lever, with the time horizon and the stock/bond mix close behind. Four sample profiles, run at the default seed:

Profile	Savings	Spending	Mix	Retire to	Withdrawal	Success
Comfortable	\$1.20M	\$45k	60/40	65-95	3.8%	90%
Baseline 4%	\$1.00M	\$40k	60/40	65-95	4.0%	86%
Conservative	\$0.90M	\$38k	50/50	67-92	4.2%	93%
Tight	\$0.70M	\$45k	55/45	64-95	6.4%	27%



Baseline 4% case. Bands show where the balance lands across 10,000 runs (10th-90th and 25th-75th percentiles); the line is the median.

Validation

The engine ships with a 7-test suite. The headline check turns volatility off and confirms the simulated path equals the closed-form annuity formula exactly, to 9 decimal places, which proves the arithmetic rather than just its plausibility. The rest confirm the result moves the right way with spending, savings, and horizon; reproduces under a fixed seed; stays stable as the run count grows; and lands the 4% case in the published band. The Python engine and the browser version use the same formulas and agree to simulation tolerance.

Limitations

Yearly returns are independent and identically distributed; real markets have fat tails and some autocorrelation this does not model. Spending is constant in real terms. There is no Social Security, pension, tax, or fee modeling. Stocks and bonds are assumed uncorrelated. Long-run assumptions are not a forecast, and any 30-year stretch can fall outside them. This is a teaching and demonstration tool, not a financial plan.

Compliance note

This is a personal prototype that uses hypothetical illustrations and carries a clear 'not financial advice' disclaimer. Putting a tool like this on a regulated firm's website, for example a firm under FINRA/SEC oversight, would require the firm's own compliance review of the assumptions, disclosures, and any lead capture.